

CHAPTER 9

REACTION

RATES

By the end of this chapter you will have covered the following material.

Science Understanding

- Varying the conditions present during chemical reactions can affect the rate of the reaction and in some cases the identity of the products ([ACSCH068](#))
- The rate of chemical reactions can be quantified by measuring the rate of formation of products or the depletion of reactants ([ACSCH069](#))
- Collision theory can be used to explain and predict the effect of concentration, temperature, pressure and surface area on the rate of chemical reactions by considering the structure of the reactants and the energy of particles ([ACSCH070](#))
- The activation energy is the minimum energy required for a chemical reaction to occur and is related to the strength of the existing chemical bonds; the magnitude of the activation energy influences the rate of a chemical reaction ([ACSCH071](#))
- Energy profile diagrams can be used to represent the enthalpy changes and activation energy associated with a chemical reaction ([ACSCH072](#))
- Catalysts, including enzymes and metal nanoparticles, affect the rate of certain reactions by providing an alternative reaction pathway with a reduced activation energy, hence increasing the proportion of collisions that lead to a chemical change ([ACSCH073](#))



9.1 The rate of a reaction

Just as some people run faster than others, some chemical reactions go faster than others. As scientists' knowledge of what happens during a reaction has deepened, they have been able to apply this knowledge to control the speed, or rate, of a reaction to suit our purposes.

EXPERIMENT 9.1

OBSERVING RATES OF REACTIONS

Aim

To observe the rates at which different reactions occur

Materials

- Zinc granule
- 10 mL of 1 mol L⁻¹ hydrochloric acid (HCl)
- 2 mL of 1 mol L⁻¹ copper(II) nitrate (Cu(NO₃)₂)
- 2 mL of 1 mol L⁻¹ sodium carbonate (Na₂CO₃)
- 2 test tubes

What are the risks in doing this experiment?	How can you manage these risks to stay safe?
Chemicals could splash into your eyes.	Wear safety glasses at all times.
1 mol L ⁻¹ HCl is corrosive to skin and clothing.	Wear safety glasses and protective clothing. Take care when pouring and clean up spills immediately. If hydrochloric acid is spilt on your skin, wash the affected area with plenty of water and notify your teacher.

In your write-up, add any more risks you can think of, as well as ways to manage them.

Procedure

- 1 Place the zinc granule in a test tube.
- 2 Cover the zinc with hydrochloric acid.
- 3 Record your observations for the reaction.
- 4 Place 2 mL of copper(II) nitrate in the second test tube.
- 5 Add 2 mL of sodium carbonate to the second test tube.
- 6 Record your observations of the second reaction.

Results

Present your observations in a simple table.

Analysis of results

- 1 How could you tell that a reaction had occurred in each of the test tubes?
- 2 Which reaction was quicker? How did you know?
- 3 How could you measure how quickly each reaction occurred?

Discussion

- 1 What other reactions do you know that occur quickly?
- 2 What other reactions do you know that occur slowly?

What is a rate?

Rate is how quickly one quantity changes compared to another quantity. The rate at which a car moves is how many kilometres the car travels in an hour. The rate at which someone loses weight would be the weight loss (often in kilograms) per week. Another common rate is rate of pay, which describes how much someone is paid per hour. In terms of reactions, the rate is how quickly a reaction occurs.



Alamy/Neil Tyler

▲ **Figure C9.1**
These animals are travelling at different rates. How can you tell?



◀ **Figure C9.2**
Two beakers with the same reactants. Which one is reacting at the faster rate? How do you know?

ACTIVITY 9.1

RATES IN OUR LIVES

Aim

To become aware of rates that affect our lives

What to do

- 1 Consider what you do in one day.
- 2 Make a list of anything you use, do or make for which the rate is important. For example, when you make a cup of tea, the rate at which the water heats up affects how long you have to wait for the water to boil.
- 3 For each item on your list, state whether a fast or slow rate is desirable.
- 4 If possible, make a note of how you are able to affect the rate of each item.



To see how reaction rates affect our bodies, refer to Context 4, 'Making reactions work for us', page 85.



Mark Fergus Photography

Figure C9.3 ▲

Which sugar would dissolve the fastest? How could you measure it to accurately compare the three? Try this to see if your prediction is correct.

Measuring the rate of a reaction

To calculate any rate, divide the change in one quantity by the change in the other quantity.

$$\text{Rate} = \frac{\text{change in quantity 1}}{\text{change in quantity 2}}$$

For example, at the 2012 London Olympic Games, Usain Bolt won the 200-metre sprint in 19.26 seconds. Therefore, his rate of running, or speed, was:

$$\text{Rate} = \frac{200}{19.26} = 10.4 \text{ m s}^{-1}$$

To measure the rate of a reaction, you need something measurable to change, such as:

- mass
- colour
- volume
- pH
- concentration.

If you can measure this quantity at different times during the reaction, then you can determine how quickly it is changing during the reaction. For example, in the reaction between solid calcium carbonate (CaCO_3) and a solution of hydrochloric acid (HCl), the mass of the solid will decrease as it reacts and forms the products calcium chloride (CaCl_2), water (H_2O) and carbon dioxide (CO_2), according to the equation:

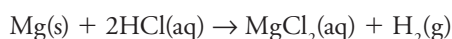


If you measure the mass of the solid at the start of the reaction and end of the reaction, and measure the time it takes for the reaction to occur, then you can calculate the rate of the reaction. For example if 2.00 g of calcium carbonate takes 48 seconds to completely react, the rate would be:

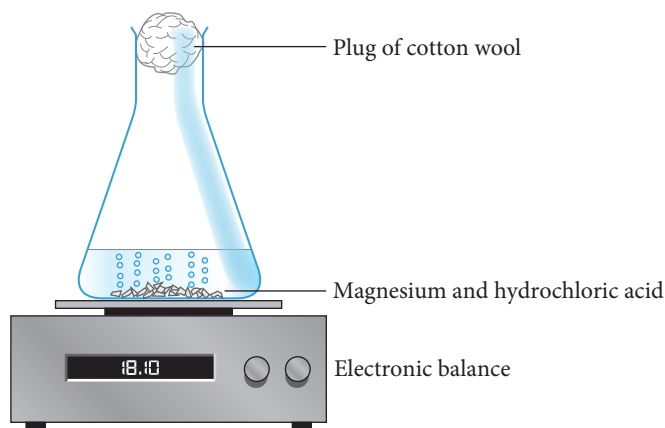
$$\text{Rate of reaction} = \frac{\text{change in mass}}{\text{change in time}} = \frac{2.00}{48} = 4.17 \times 10^{-2} \text{ g s}^{-1}$$

However, the rate does not have to be for the overall reaction. You can also calculate the rate at various stages of the reaction by taking measurements throughout the reaction time. These may then be used to calculate the rate directly, or you can graph the changes that occur during the reaction.

Consider the reaction between magnesium and hydrochloric acid:



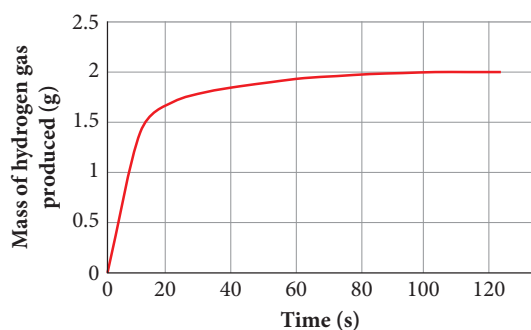
The mass of the hydrogen gas produced during the reaction is calculated by the mass lost from a flask. This is shown in Table C9.1.



◀ **Figure C9.4** Measuring the mass of hydrogen gas lost during the reaction

Table C9.1 Mass during the reaction between magnesium and hydrochloric acid

Time (s)	Mass of flask and contents (g)	Mass of hydrogen gas released (g)
0	20.00	0.00
10	18.60	1.40
20	18.30	1.70
30	18.20	1.80
40	18.15	1.85
50	18.10	1.90
60	18.06	1.94
70	18.04	1.96
80	18.02	1.98
90	18.01	1.99
100	18.00	2.00
110	18.00	2.00
120	18.00	2.00



▲ **Figure C9.5**
Mass of hydrogen gas produced

On a graph showing the changes during a reaction, the slope or **gradient** represents the rate of a reaction because it shows how quickly the reactants or products change. The graph in Figure C9.5 shows that rate of the reaction was the greatest at the start of the reaction and gradually slowed to zero at the end of the reaction. This is supported by the raw data, which shows that the changes in mass are greatest at the start of the reaction. You will learn more about why this happens later in this chapter.



GRAPHING THE CHANGES DURING A REACTION

Use this simulation to observe a reaction and the shape of the graphs produced.

EXPERIMENT 9.2

MEASURING THE RATE OF A REACTION

The rate of a reaction is determined by measuring how much reactant is consumed per unit of time, or how much product is made per unit of time.

The reaction between calcium carbonate and hydrochloric acid produces carbon dioxide, water and calcium chloride.



By measuring the amount of carbon dioxide produced at different time intervals, you can determine the rate of the reaction.

Aim

To observe and calculate the rate of reaction between calcium carbonate and hydrochloric acid at various stages

Materials

- 2 g of CaCO_3 powder
- 250 mL of 0.25 mol L^{-1} HCl
- 100 mL beaker
- 500 mL conical flask
- Spatula
- 250 mL measuring cylinder
- Paper
- Graph paper
- Electronic balance
- Stopwatch
- Calculator

What are the risks in doing this experiment?	How can you manage these risks to stay safe?
Chemicals could splash in your eyes.	Wear safety glasses at all times.
0.25 mol L^{-1} HCl is corrosive to skin and clothing.	Wear safety glasses and protective clothing. Take care when pouring and clean up spills immediately. If hydrochloric acid is spilt on your skin, wash the affected area with plenty of water and notify your teacher.

In your write-up, add any more risks you can think of, as well as ways to manage them.

Procedure

- 1 Measure 2 g of CaCO_3 powder into the clean, dry beaker.
- 2 Accurately measure 250 mL of 0.25 mol L^{-1} HCl.
- 3 Place the HCl in a clean conical flask and place this on the electronic balance. Record the mass of the flask and acid.
- 4 Calculate the mass of the flask, HCl and CaCO_3 by adding your values from steps 1 and 3.
- 5 Place the CaCO_3 on a clean piece of paper, and roll the paper so that it can easily be inserted into the top of the flask.
- 6 Add the CaCO_3 to the flask (while it is on the electronic balance) and immediately start the stopwatch and record the mass.
- 7 Record the mass (and time) as frequently as you can.
- 8 When the reaction is complete, take the flask off the electronic balance.

Results

Organise your data into an appropriate table, remembering to include a column to calculate the cumulative mass of carbon dioxide gas produced for each time.

Analysis of results

- 1 What happened to the mass of the flask and its contents? Why did this occur?
- 2 How can you use the mass of the flask to determine the mass of the CO_2 produced?

- 3 Calculate the cumulative mass of the CO_2 produced for each time.
- 4 Graph the mass of CO_2 produced against time.
- 5 When was the rate of reaction the fastest? How could you know this from the graph?
- 6 When was the rate of reaction the slowest? How could you know this from the graph?
- 7 At what time was the reaction complete? How could you know this from the graph?
- 8 Calculate the average rate of reaction (average rate = $\frac{\text{total change in mass}}{\text{time for mass change}}$).
- 9 Draw tangents to the graph at three different times and calculate the gradients (or slopes) of the lines. These are the rates of the reaction at those times. Ask your teacher if you do not know how to draw a tangent.



DRAWING A TANGENT TO A CURVE

Visit this website to see how to draw a tangent to a curve.

Discussion

- 1 What is the difference between qualitative data and quantitative data?
- 2 During the experiment, how could you observe the rate of the reaction?
- 3 Did your quantitative data support your qualitative data regarding the rate of the reaction?
- 4 What aspects of the experiment may have reduced the accuracy of the data?
- 5 How could you improve the experiment to ensure that the results were more accurate and reliable?

Conclusion

What conclusion can you make about the rate of the reaction at various stages of a reaction?

QUESTION SET 9.1

Remembering

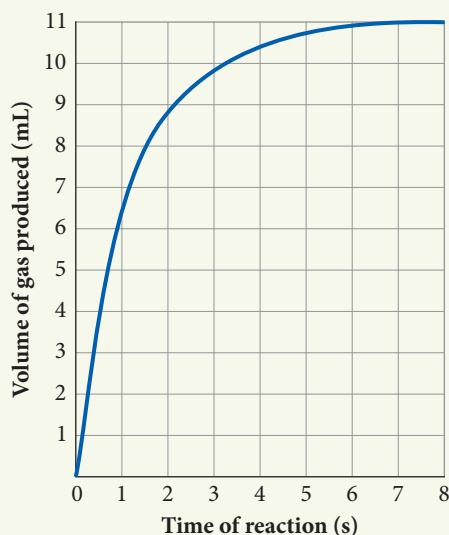
- 1 Define 'rate'.
- 2 What data would you record to be able to calculate or compare the rates of different reactions?
- 3 During a reaction, when is the rate the fastest?

Understanding

- 4 Describe how to calculate the rate of a reaction.
- 5 Describe how the slope (or gradient) of a graph indicates the rate.

Applying

- 6 During a reaction 20 grams of product is formed in 4 minutes. Calculate the rate of the reaction.
- 7 Copy the graph in Figure C9.6 and draw tangents to the curve at times 0, 1 and 5 to calculate the rate of reactions at each of these times.



◀ **Figure C9.6**
Production of carbon dioxide gas during the decomposition of copper(II) carbonate

9.2 What is needed for a reaction

To be able to understand why some reactions occur faster than others, you must first understand what happens to the particles during a reaction.

ACTIVITY 9.2

MODELLING A REACTION

Aim

To use molecular models to represent a chemical reaction

You will need

- Molecular model kit; for example, Molymod

What to do

- 1 Using the kit, make two molecules of hydrogen gas (H_2) and one molecule of oxygen gas (O_2) as shown in Figure C9.7.
- 2 Use these molecules to make two molecules of water (H_2O).
- 3 Describe what had to happen for the reactants (hydrogen gas and oxygen gas) to become the product (water).

What did you discover?

Based on the reaction you modelled, what has to happen for the reactants in a chemical reaction to become products?

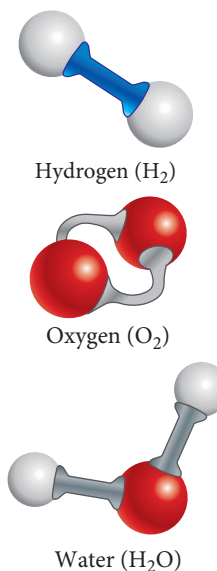


Figure C9.7 ▲
Modelling a chemical reaction

ACTIVITY 9.3

DODGEM CAR CRASH ANALOGY

In this activity you are going to compare a 'successful' dodgem car crash with a chemical reaction.

Aim

To compare the components of a chemical reaction with those of dodgem cars when they crash

What to do

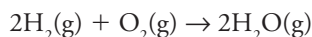
- 1 In small groups, brainstorm the following ideas.
 - Consider a dodgem car crash.
 - What would make a dodgem car crash 'successful'?
 - Are all dodgem car crashes successful?
 - What has to happen during a dodgem car crash for it to be successful?
- 2 Write a paragraph describing a dodgem car crash and what has to happen for it to be successful.
- 3 Save your paragraph to refer to later.



Figure C9.8 ▲
Is this a successful dodgem car crash?

Alamy/Art Directors & Trip

Consider the reaction between hydrogen gas and oxygen gas that occurs in a hydrogen fuel cell and the 'pop test'.



For this reaction to occur:

- bonds in hydrogen molecules must break to separate the hydrogen atoms
- bonds in oxygen molecules must break to separate the oxygen atoms
- new bonds must form between the hydrogen and oxygen atoms to make the water molecules.

This reaction requires enough energy to overcome the forces of attraction holding the atoms of the reactants in their original form. So energy is required to break the covalent bonds within the hydrogen molecules and oxygen molecules.

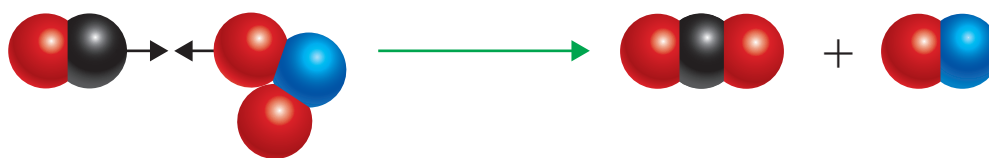
This reaction also requires the appropriate atoms to come into contact with each other so that the new bonds of the products can form. This means that two hydrogen atoms must come in contact with one oxygen atom to be able to form a water molecule.

Collision theory

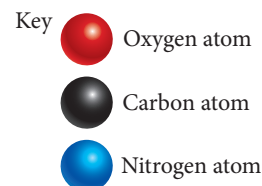
To understand why reactions occur quickly or slowly, you need to know what happens during a chemical reaction. **Collision theory** is a model that allows us to explain what happens during a chemical reaction and the factors that affect it. This theory states that, for a reaction to occur, the reactants must collide with sufficient energy and in an appropriate orientation. Collisions between reactant particles can be classified as either successful or unsuccessful. A successful collision has sufficient energy and an appropriate orientation to allow old bonds to be broken and new bonds to be formed. An unsuccessful reaction is one in which the energy and/or orientation are not satisfactory. The more successful collisions there are in a given time, the faster the rate of the reaction because more reactants are forming products.

Collision theory states that, for a reaction to occur, the particles must collide with sufficient energy and in the appropriate orientation.

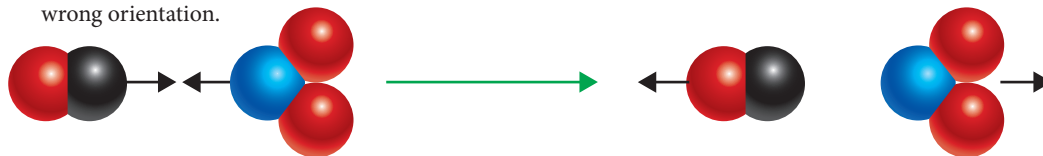
Particles are colliding at the right orientation and with sufficient energy.



Successful collision



Particles are colliding at the wrong orientation.



Unsuccessful collision

Particles have bounced off each other; speeds and direction of movement have changed.

▲ Figure C9.9

The correct orientation is needed for a successful collision.

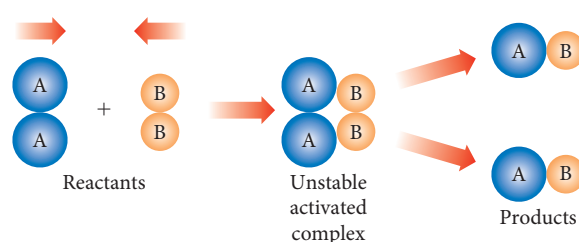
ACTIVITY 9.4

DODGEM CAR CRASH ANALOGY AND COLLISION THEORY

- 1 Review the dodgem car crash analogy in Activity 9.3.
- 2 In a dodgem car crash, what would be the equivalent of these parts of the collision theory?
 - a The collision
 - b Enough energy
 - c Appropriate orientation
- 3 For each part of Question 2, explain the relevance to a 'successful' dodgem car crash.

During the progress of a reaction, as reactants form the products, there exists a brief moment where the bonds of the reactants have broken, but the bonds of the products are not yet formed. This produces a temporary, highly unstable structure called the **activated complex** or **transition state**. As energy was required to break the bonds of the reactants, the activated complex has more energy than either the reactants or products.

Figure C9.10 ►
Stages of a reaction
($A_2 + B_2 \rightarrow 2AB$)



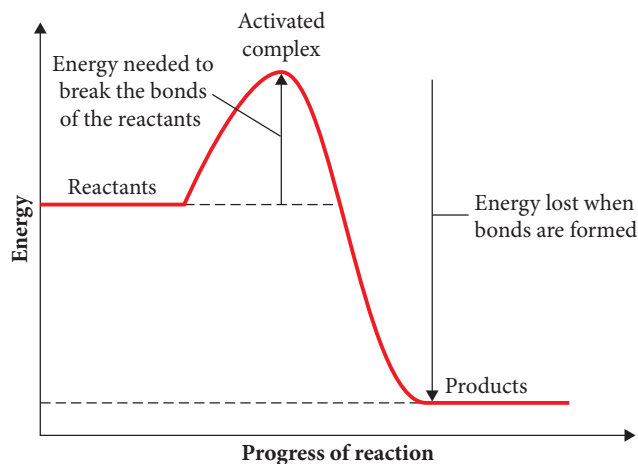
The activated complex will quickly form new bonds to produce the products, releasing energy. This means that the atoms that will bond need to be positioned next to each other so they can form the new bonds. This is why the orientation of the colliding particles needs to be correct.

Energy profile diagrams

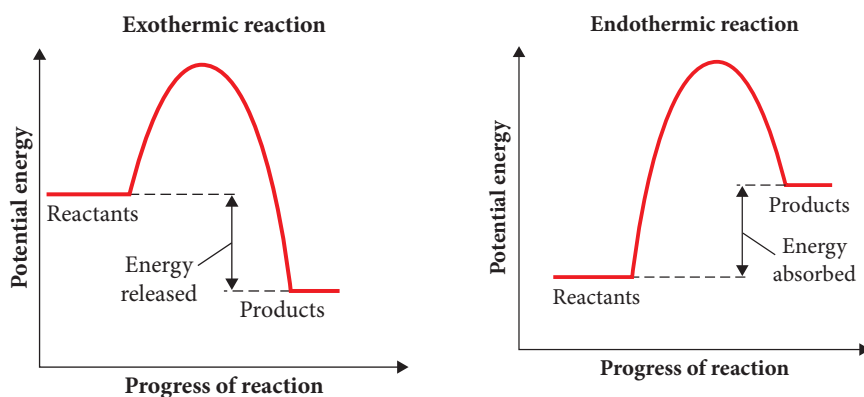
The total energy possessed by a chemical substance is called **enthalpy** or heat content, and is given the symbol H . In a reaction, the reactants, activated complex and products each have their own enthalpies. These can be represented in an **energy profile diagram**.

Energy is needed to break the bonds of the reactants; therefore, the activated complex has more energy than the reactants. Similarly, energy is lost when the bonds of the products are formed. Therefore, the activated complex has more energy than the products as well.

Figure C9.11 ►
An energy profile diagram



If more energy is lost when the products are formed than was gained to form the activated complex, then the difference in energy is lost from the system to the surroundings. This occurs in an **exothermic reaction**. Conversely, if more energy is gained than is lost, the excess must be gained from the surroundings. This occurs in an **endothermic reaction**.



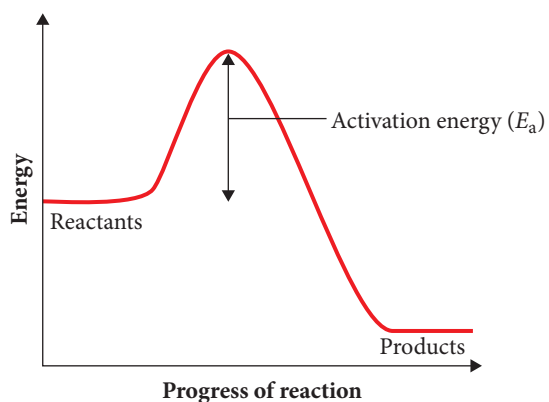
◀ **Figure C9.12**
Energy profile diagrams for endothermic and exothermic reactions

To revise exothermic and endothermic reactions, see Chemistry section 4.3 on page 213.

Activation energy

The amount of energy needed to form the activated complex is called the **activation energy** and is often represented as E_a . As this energy is needed for the bonds of the reactants to break, the activation energy is the minimum energy required for a reaction to occur. The activation energy can be determined from the difference in energy between the activated complex and the reactants.

The activation energy is the minimum energy required for a chemical reaction to occur and influences the rate of a chemical reaction.



◀ **Figure C9.13**
Activation energy

At a given temperature, particles will have different amounts of **kinetic energy**, with the average kinetic energy determining the temperature. This is represented in the **Maxwell-Boltzmann distribution**, which graphically shows the energies of the particles at a particular temperature (Figure C9.14).

From Figure C9.13, you can see that not all particles have the same amount of energy. Only those particles with the activation energy, or greater, will have enough energy to break the bonds of the reactants and have a successful collision. On the graph, the proportion of particles with that amount of energy is represented by the area under the curve above the activation energy (Figure C9.15 on page 378).



EFFECT OF ENERGY ON COLLISIONS

Use this collision simulation to see what happens if the particles in a reaction have or don't have enough energy.



EFFECT OF ENERGY AND ORIENTATION ON COLLISIONS

Use this simulation to see the effect of energy and orientation on reactions.

Figure C9.14 ►

The Maxwell-Boltzmann distribution shows the energy of molecules at a particular temperature.

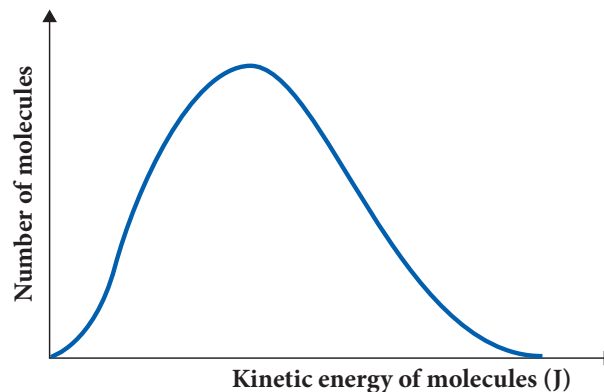
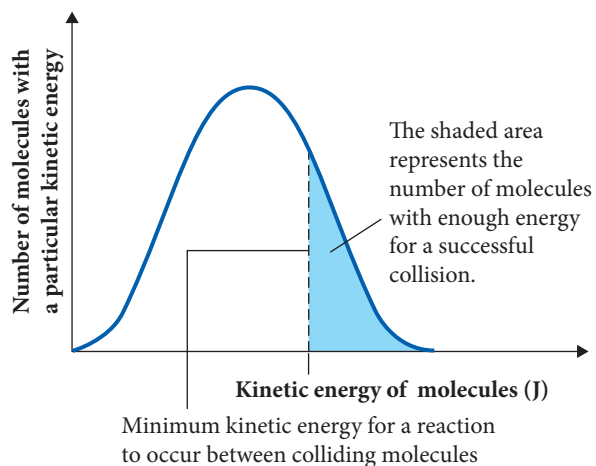


Figure C9.15 ►

The number of molecules with enough energy for a successful collision



To see applications of collision theory on reactions, refer to Context 4, 'Making reactions work for us', page 85.



QUESTION SET 9.2

Remembering

- 1 State the collision theory.
- 2 Define:
 - a activation energy.
 - b activated complex.
- 3 Draw a Maxwell-Boltzmann distribution.

Understanding

- 4 Explain why a successful collision requires:
 - a sufficient energy.
 - b the correct orientation.
- 5 What will happen to two particles that collide with energy less than the activation energy?

Applying

- 6 Draw and label an energy profile diagram for an:
 - a endothermic reaction with an activation energy of 1000 J.
 - b exothermic reaction with an activation energy of 200 J.

9.3 What affects rate

Collision theory tells us that the rate of a reaction depends on the number of successful collisions between reactant particles in a given time. Therefore, anything that increases the number of successful collisions will increase the rate of the reaction.

The number of successful collisions depends on the:

- total number of collisions
- percentage of collisions that are successful.

Any factor that will increase either of these factors will lead to an increase in the reaction rate. Conversely, anything that decreases these factors will lead to a decrease in the reaction rate.

For example, if there are 1000 collisions between reactants per second and 60% are successful, there will be 600 successful collisions per second. If the number of collisions is increased to 2000 per second, even if only 60% are successful, there will now be 1200 successful collisions per second and therefore the reaction rate has increased. Alternatively, if there are still 1000 collisions, but 80% of them are successful, there are now 800 successful collisions per second and so the reaction rate has still increased.

Table C9.2 Successful collisions

Number of collisions	Percentage of collisions that are successful	Number of successful collisions
1000	60	600
1000	80	800
2000	60	1200
2000	80	1600

INVESTIGATION 9.1

CHANGING THE COLOUR OF POTASSIUM PERMANGANATE

Rhubarb contains oxalic acid, which will react with acidified potassium permanganate, causing it to decolorise. This can be represented by the following equation.



The time for this change in colour to occur can be measured and used to calculate the rate of the reaction.

The rhubarb can be fresh or frozen rhubarb sticks, or in the form of a puree that can be made by gently boiling rhubarb in water.

Your task is to consider all the factors that may affect the time for the permanganate to decolorise. Choose one of these factors and design an investigation to determine its effect on rate of the reaction.

What is your aim?

Write down an aim for your investigation, stating clearly what you are investigating.

What will you need?

Make a list of all the materials and equipment that you will need to carry out your investigation.

Make sure that you are specific about the type, number and size of all the materials and apparatus needed.

What are the risks?

Construct a table similar to the one on page 380. Identify specific risks involved in the investigation and ways that you will manage the risks to avoid injuries or damage to equipment. Ask your teacher to check your risk assessment before you proceed.



REACTING WITH RHUBARB

Visit this website to find out more about the reaction between oxalic acid in rhubarb and potassium permanganate.



HEALTH AND SAFETY NOTES FOR RHUBARB REACTION

Check your risk assessment for the rhubarb investigation by visiting this website.

What are the risks in doing this investigation?	How can you manage these risks to stay safe?

How will you carry out your investigation?

Clearly write down, in numbered steps, how you are going to carry out your investigation to achieve your aim. Make sure your steps clearly show how you are going to control or measure all your variables.

What results will you collect?

- 1 What is your independent variable? How will you change this?
- 2 What is your dependent variable? How will you measure this?
- 3 Is there other data that you need to measure and record?
- 4 Will you collect quantitative and/or qualitative results? State what sort of results you will be collecting and when you will collect them.

How will you analyse your results?

- 1 How will you calculate the rate of reaction from your data?
- 2 State how you analysed your results and write the analysis here.

What have you found?

What did you find when you analysed your results?

What do you conclude?

- 1 How can science be used to explain your conclusion?
- 2 Write a conclusion to your investigation based on your aim and your results.

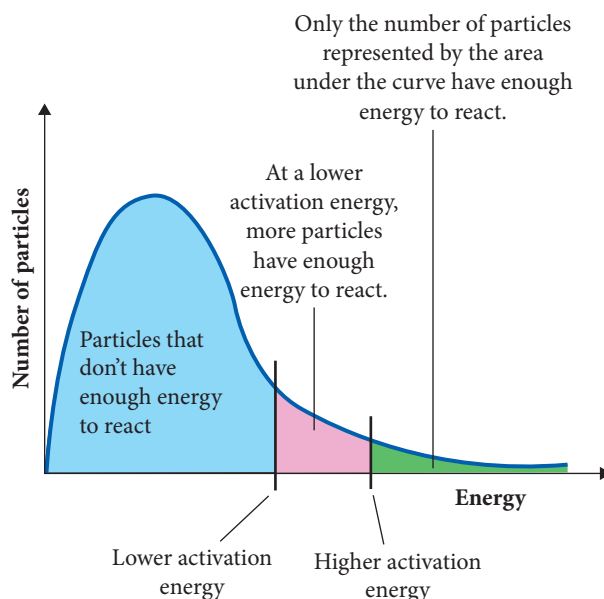
Ideas for improvement

Did you experience any problems with your materials or method? How would you overcome this if you were to repeat this investigation?

Taking it further

Could you undertake any further investigation to find out more information about this topic? If yes, what would you want to find out, and how would you go about doing that?

Figure C9.16 ►
The number of successful collisions with different activation energies



Nature of reactants

The nature of the reactants refers to what the reactants are and what has to happen for them to form the products.

If the bonds of the reactants that need to be broken are weak or few, then they will only require a small amount of energy to break. This means that the activation energy will be low, and a larger number of particles will have enough energy for a successful collision.

The size of the particles may also affect the rate of the reaction. If the particles are larger, then there is a greater chance that they will collide.

ACTIVITY 9.5

DODGEM CAR CRASH ANALOGY AND NATURE OF REACTANTS

What to do

- 1 Review the dodgem car crash analogy in Activity 9.3.
- 2 What would be the equivalent of the nature of reactants for your dodgem car crash analogy?
- 3 How could you increase the 'activation energy' for your dodgem car crash?
- 4 How could you decrease the 'activation energy' for your dodgem car crash?
- 5 Are these factors that real car manufacturers are able to consider to make cars safer?

What have you discovered?

Write a paragraph summarising how the nature of reactants will affect the rate of dodgem car crashes.



EFFECT OF THE NATURE OF REACTANTS ON REACTION RATE

View this animation to see the effect of reactant size on reaction rate.

Concentration and pressure

Concentration refers to the number of particles in a given volume. The **pressure** of a gas is determined by the concentration of the gaseous particles, and therefore you can consider the two factors simultaneously. The number of particles in a given volume will determine the number of collisions that occur. The greater the concentration, or pressure, the greater the number of particles and therefore the greater the number of collisions. Even though the percentage of collisions that will be successful will remain the same, there will be more successful collisions in total because there are more collisions. This means that as the concentration, or pressure, of the reactants increases, the reaction rate increases.



EFFECT OF CONCENTRATION ON COLLISIONS

View this animation to compare the number of collisions between acids and a marble chip in different solutions. Which one would have the fastest reaction rate? Choose 'View online', then click the green arrow to start the animation.



EFFECT OF PRESSURE ON REACTION RATE

View this animation of collision theory to see the effect of pressure on the rate of reaction.



Figure C9.17 Solutions of different concentrations



ACTIVITY 9.6

SCREAMING GUMMI BEAR

Aim

To observe a fast reaction and investigate the factors that contribute to its speed

What you will need:

- computer with internet access

What to do

- 1 Watch the video of the experiment at the weblink 'Screaming gummi bear'. This is an example of a very fast reaction. What observations can you make that indicate the rate of this reaction?
- 2 Research to find the equation for the reaction that is occurring.
- 3 Research the reaction to find out why it occurs so quickly.
- 4 Your teacher may demonstrate this experiment to you. How was it similar to the video? How was it different? Why do you think it may have been different? What did you discover?
- 5 What factors affected the rate of the Screaming gummi bear experiment?



SCREAMING GUMMI BEAR

Watch this video of the screaming gummi bear experiment.

EXPERIMENT 9.3

EFFECT OF CONCENTRATION ON RATE OF REACTION

Sodium thiosulfate reacts with dilute hydrochloric acid to produce sodium chloride, sulfur, sulfur dioxide and water.



When the concentration of sulfur is high enough, the solution will become cloudy. The time for this to occur can be measured to indicate the rate of the reaction.

Aim

To investigate the effect of concentration on the rate of the reaction between sodium thiosulfate and dilute hydrochloric acid

Materials

Per trial:

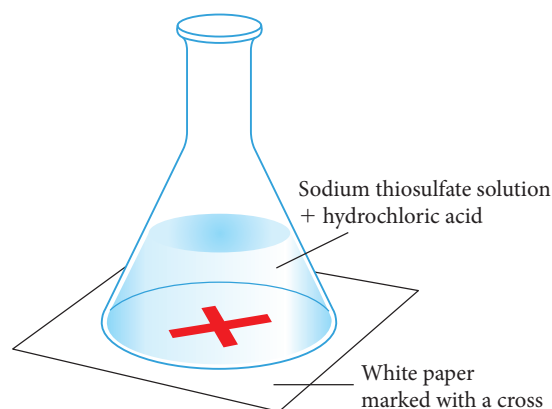
- 45 mL of 0.25 mol L⁻¹ sodium thiosulfate (Na₂S₂O₃)
- 45 mL of 0.125 mol L⁻¹ sodium thiosulfate (Na₂S₂O₃)
- 45 mL of 0.0625 mol L⁻¹ sodium thiosulfate (Na₂S₂O₃)
- 15 mL of 2 mol L⁻¹ hydrochloric acid (HCl)
- Stopwatch
- 3 × 100 mL conical flask
- 1 × 50 mL and 1 × 10 mL measuring cylinder
- White paper
- Pen or marker pen

What are the risks in doing this experiment?	How can you manage these risks to stay safe?
Chemicals could splash in your eyes.	Wear safety glasses at all times.
2 mol L ⁻¹ HCl is corrosive to skin and clothes.	Wear gloves and an apron.
SO ₂ gas irritates the nose, throat and airways.	Conduct the experiment in a well-ventilated area and don't breathe in fumes.

In your write-up, add any more risks you can think of, as well as ways to manage them. In particular, expand on the three risks listed to identify specific risks involved with each of them.

Procedure

- 1 Draw a large cross in the centre of the piece of white paper.
- 2 Place the conical flask over the cross.
- 3 Accurately measure 45 mL of $0.0625 \text{ mol L}^{-1} \text{ Na}_2\text{S}_2\text{O}_3$ in a measuring cylinder.
- 4 Pour the $\text{Na}_2\text{S}_2\text{O}_3$ into the conical flask.
- 5 Accurately measure 5 mL of $2 \text{ mol L}^{-1} \text{ HCl}$ in a measuring cylinder.
- 6 Have a stopwatch ready to start timing.
- 7 Pour the HCl into the conical flask, starting the stopwatch immediately.
- 8 Quickly mix the two solutions by carefully swirling the conical flask.
- 9 Place the conical flask onto the cross and record the time it takes before you can no longer see the cross when looking from above. (Be careful not to breathe in any fumes from the reaction.)
- 10 Repeat steps 3–9 for the other two solutions.
- 11 If time and resources allow, repeat steps 3–10 for repeat trials.
- 12 Place your solutions in the waste beaker, and rinse the conical flask.



▲ **Figure C9.18**
Reaction between sodium thiosulfate and dilute hydrochloric acid

Results

Enter your results in a results table like the one below.

Concentration of sodium thiosulfate (mol L^{-1})	Time taken for the cross to disappear (s)			
	Trial 1	Trial 2	Trial 3	Average
0.0625				
0.125				
0.250				

Analysis of results

- 1 Graph the average time taken for the cross to disappear against the concentration of sodium thiosulfate.
- 2 How is the time taken for the cross to disappear related to the rate of the reaction?
- 3 What happened to the time taken for the cross to disappear as the concentration of sodium thiosulfate increased?
- 4 What is the relationship between the concentration of sodium thiosulfate and the rate of the reaction?
- 5 Does this support the theoretical trend?
- 6 Were there any anomalies? If yes, why do you think they occurred?

Discussion

- 1 Use collision theory to explain the relationship between concentration and the rate of the reaction.
- 2 Where might errors have occurred in the method that would have affected the accuracy of the results?
- 3 How could the method be improved to increase the accuracy and reliability of the results?

Conclusion

What conclusion can you make about the concentration of reactants and the rate of the reaction?

Taking it further

- 1 How could you further investigate the effect of concentration on the rate of the reaction in this reaction?
- 2 How could you further investigate the effect of concentration on the rate of reactions in general?



EFFECT OF CONCENTRATION ON REACTION RATE

View this animation of collision theory to see the effect of concentration on the rate of reaction.

Changing the volume may also change the concentration. In a solution, adding water will dilute the solution, thereby decreasing the concentration. Alternatively, evaporating some of the water will decrease the volume and increase the concentration. In a gas, if the volume of the container is increased, then the concentration of all the gases will decrease. Alternatively, if the volume of the container is decreased, then the concentration of all the gases will increase.

During the course of a reaction, reactants are being used up to form the product. This means that the concentration of the reactants is decreasing. From our understanding of the effect of concentration on the rate of a reaction, it makes sense that the rate of the reaction will decrease as time progresses during the reaction. On a graph of the reactants or products versus time, the rate of the reaction is indicated by the slope of the graph. Therefore, the slope will be steepest at the start of the reaction and will gradually decrease to zero at the completion of the reaction. This can be seen clearly in Figure C9.5 on page 371.

ACTIVITY 9.7

DODGEM CAR CRASH ANALOGY – CONCENTRATION

Aim

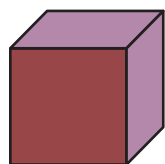
To compare the effect of concentration on the rate of a reaction with dodgem car crashes

What to do

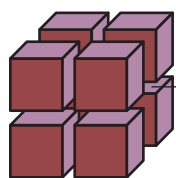
- 1 Review the dodgem car crash analogy in Activity 9.3.
- 2 In your dodgem car crash analogy what would be the equivalent to concentration?
- 3 How would changing this concentration equivalent affect the rate of dodgem car crashes?
- 4 Would it matter if only one 'reactant' or all 'reactants' were changed?

What did you discover?

Write a paragraph summarising how changes in concentration will affect the rate of dodgem car crashes.



Low surface area



High surface area

Inner particles are now exposed and are available for a collision.

Surface area

For a reaction to occur, the reactant particles must be able to collide. When one of the reactants is a solid, only the particles on the surface of the solid are available for collision. Once these have reacted, then the particles underneath are exposed and are able to react, and so on.

If the solid is divided or crushed, then the inner particles become exposed and, although the amount of reactant remains the same, the total **surface area** has increased. This means that more of the solid is available for collisions and is able to react. Therefore, there are more collisions and thus the reaction rate will increase.

We naturally use this method of increasing the rate of a reaction every day. It is why we chop wood into small pieces of kindling to start the fire, why we cut potatoes up into small pieces to cook them quickly and why we use granulated sugar instead of sugar cubes in our coffee.

Figure C9.19 ▲

Dividing a solid will increase the surface area exposed for a reaction.

EXPERIMENT 9.4

EFFECT OF SURFACE AREA ON RATE OF REACTION

Calcium carbonate will react with dilute hydrochloric acid according to the following equation:



During the reaction, bubbles of carbon dioxide can be seen. Therefore, when there are no more bubbles being produced, the reaction has stopped. The time for this to occur can be measured to compare the rates of reactions.

In this experiment, you will change the surface area of a fixed mass of solid calcium carbonate. By comparing the time taken for the reactions to occur, you will be able to determine the effect of surface area on the rate of the reaction.

Aim

To determine the effect of surface area on the rate of the reaction between calcium carbonate and dilute hydrochloric acid

Materials

- 3 calcium carbonate (CaCO_3) chips
- Calcium carbonate (CaCO_3) powder
- 50 mL of 2 mol L^{-1} HCl
- Electronic balance
- 10 mL measuring cylinder
- Stopwatch
- 2 large test tubes

What are the risks in doing this experiment?	How can you manage these risks to stay safe?
Chemicals may splash in your eyes.	Wear safety glasses at all times.
2 mol L^{-1} HCl is corrosive to skin and clothes.	Wear gloves and an apron.

In your write-up, add any more risks you can think of, as well as ways to manage them. In particular, expand on the two risks listed to identify specific risks involved with each of them.

Procedure

- 1 Measure 8 mL of HCl in a measuring cylinder.
- 2 Weigh and record the mass of three calcium carbonate chips.
- 3 Place the chips in one test tube.
- 4 Have a stopwatch ready to start timing.
- 5 Pour the HCl into the test tube and start the stopwatch.
- 6 Stop the stopwatch when no more bubbles are produced, and record this time.
- 7 Carefully weigh the same mass of calcium carbonate powder as that of the calcium carbonate chips.
- 8 Repeat steps 3–6 for the powder.
- 9 If time permits, repeat steps 1–8 for a number of trials.
- 10 Place all the product solutions into the waste beakers as directed by your teacher.

Results

Record your results in a results table like the one below.

Form of calcium carbonate	Time for the reaction to complete (s)			
	Trial 1	Trial 2	Trial 3	Average
Chips				
Powder				

Analysis of results

- 1 Which form of calcium carbonate took the least time to react?
- 2 Which form of calcium carbonate has the fastest rate of reaction?
- 3 Which form of calcium carbonate had the greatest surface area? Explain why.
- 4 What is the relationship between surface area and the rate of reaction?
- 5 Do your results support the theory about the effect of surface area on the rate of reaction?

Discussion

- 1 Was this a fair test? Explain why or why not.
- 2 Were the results accurate and reliable? Explain why or why not.
- 3 How could the method be improved to more fairly and accurately test the effect of surface area on the rate of reaction between calcium carbonate and dilute hydrochloric acid?

Conclusion

What conclusion can you make from this experiment?

Taking it further

How could you further investigate the effect of surface area on the rates of reactions?

A similar effect to increasing the surface area is seen when the reactants are stirred. Stirring mixes the reactants together, exposing more of the reactants to each other. This increases the number of collisions and therefore increases the reaction rate. We use this phenomenon when making a cup of tea or coffee. By stirring the sugar, we are able to increase the rate at which it dissolves.

Figure C9.20 ►
Stirring will have the same effect as increasing the surface area as more particles are exposed to one another.



Mark Ferguson Photography

ACTIVITY 9.8

DODGEM CAR CRASH ANALOGY – SURFACE AREA

Aim

To compare the effect of surface area on the rate of a reaction with dodgem car crashes

What to do

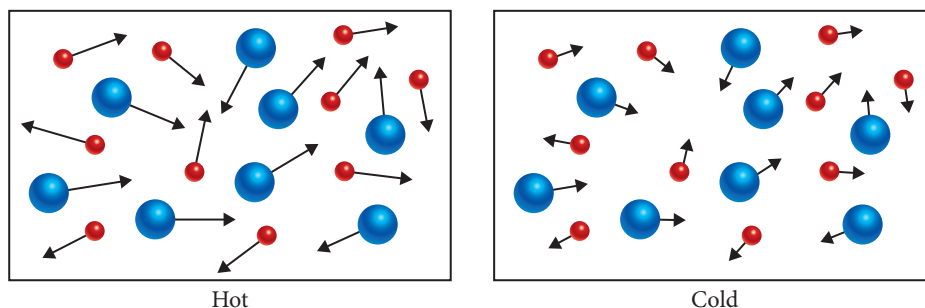
- 1 Review the dodgem car crash analogy in Activity 9.3.
- 2 In your dodgem car crash analogy, what would be the equivalent to surface area?
- 3 How would changing this surface area equivalent affect the rate of dodgem car crashes?

What did you discover?

Write a paragraph summarising how the surface area will affect the rate of dodgem car crashes.

Temperature

The **temperature** of a substance is a measure of the average kinetic energy of the particles. This energy is the energy due to movement and is dependent on the speed, or velocity, of the particles. As the temperature of a substance increases the average kinetic energy also increases, and hence the average speed of the particles also increases.



▲ **Figure C9.21**

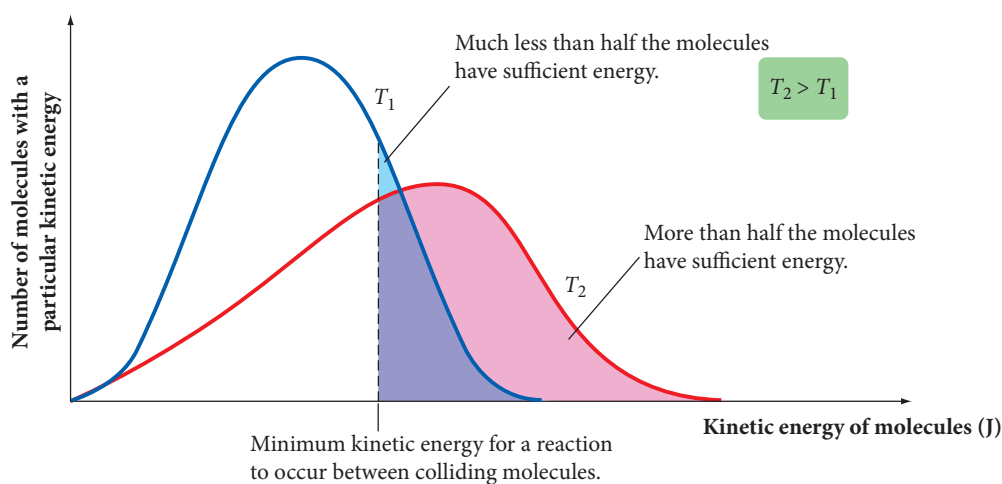
Temperature changes the speed at which particles move (as represented by the length of the arrows).

As the temperature of reactants increases there is a dual effect on the reaction rate.

1 Increased percentage of collisions that are successful

The average kinetic energy of the particles increases. This means that more particles have enough energy to overcome the activation energy, form the activated complex and become products. Therefore, a greater percentage of collisions are successful, increasing the reaction rate.

At a higher temperature the Maxwell–Boltzmann distribution (see Figure C9.22) is pushed to the right. The area under the curve to the right of the activation energy is now greater, representing a greater proportion of particles with enough energy for a successful collision.



▲ **Figure C9.22**

The Maxwell–Boltzmann distribution for different temperatures

2 Increased number of collisions

With the increase in temperature, the particles now have a greater average speed. This means that they will collide with each other more frequently, increasing the number of collisions. This will also increase the rate of the reaction. However, this effect is much less significant than the effect of the increased energy.

Kinetic energy can be calculated by using: $\text{Kinetic energy} = \frac{1}{2} \text{mass} \times \text{velocity}^2$ where velocity is the speed in a straight line. This means that if the kinetic energy of an object increases, then the velocity increases. This also means that if two objects at the same temperature (so have the same kinetic energy) have different masses, then their velocities will also be different. The object with the greater mass will have the lower velocity.



EFFECT OF TEMPERATURE ON REACTION RATE 1

View this animation to see the effect of temperature on reaction rate. Vary the temperature to see how long the reaction takes.



EFFECT OF TEMPERATURE ON REACTION RATE 2

View this animation to see the effect of temperature on reaction rate.

EXPERIMENT 9.5

EFFECT OF TEMPERATURE ON RATE OF REACTION

You have previously seen the reaction between sodium thiosulfate and dilute hydrochloric acid.



In this experiment, you will change the temperature of the solutions to investigate the effect of temperature on the rate of the reaction.

Aim

To investigate the effect of temperature on the rate of the reaction between sodium thiosulfate and dilute hydrochloric acid

Materials

Per trial:

- 150 mL of 0.25 M sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$)
- 15 mL of 2 mol L^{-1} HCl
- Crushed ice
- Hot water
- Stopwatch
- 2 $\times$ 1 L beakers (or similar container for ice bath and hot water bath)
- 3 $\times$ 100 mL conical flask
- 50 mL measuring cylinder
- 10 mL measuring cylinder
- White paper
- Pen or marker pen
- Thermometer

What are the risks in doing this experiment?	How can you manage these risks to stay safe?
Chemicals may splash in your eyes.	Wear safety glasses at all times.
2 mol L^{-1} HCl is corrosive to skin and clothes.	Wear gloves and an apron.
SO_2 gas irritates the nose, throat and airways.	Conduct the experiment in a well-ventilated area; do not breathe in fumes.

In your write-up, add any more risks you can think of, as well as ways to manage them. In particular, expand on the three risks listed to identify specific risks involved with each of them.

Procedure

- 1 Draw a large cross in the centre of a blank piece of white paper.
- 2 Accurately measure 45 mL of the 0.25 mol L^{-1} $\text{Na}_2\text{S}_2\text{O}_3$ in a measuring cylinder.
- 3 Pour the $\text{Na}_2\text{S}_2\text{O}_3$ into the first conical flask.
- 4 Place some crushed ice and a small amount of water into the 1 L beaker, and carefully place the conical flask into the ice bath. Ensure that there is enough ice to reach the height of the solution in the flask.
- 5 Repeat steps 1–4 with the second conical flask, using hot water instead of ice to make a hot water bath.
- 6 Accurately measure 45 mL of the 0.25 mol L^{-1} $\text{Na}_2\text{S}_2\text{O}_3$ in a measuring cylinder.
- 7 Pour the $\text{Na}_2\text{S}_2\text{O}_3$ into the third conical flask.
- 8 Measure and record the temperature of the $\text{Na}_2\text{S}_2\text{O}_3$ in all three conical flasks.
- 9 Accurately measure 5 mL of the 2 mol L^{-1} HCl in a measuring cylinder.
- 10 Have a stopwatch ready to start timing.
- 11 Pour the HCl into the first conical flask, starting the stopwatch immediately.
- 12 Quickly mix the two solutions by carefully swirling the conical flask.

- 13 Place the conical flask onto the cross and record the time it takes before you are no longer able to see the cross when looking from above. (Be careful not to breathe in any fumes from the reaction).
- 14 Repeat steps 9–13 for the other two solutions.
- 15 If time and resources allow, repeat steps 2–13 for repeat trials.
- 16 Place your solutions in the waste beaker, and rinse the conical flask.

Results

Record your results in a results table like the one below.

Temperature of sodium thiosulfate (°C)	Time taken for the cross to disappear (s)			
	Trial 1	Trial 2	Trial 3	Average

Analysis of results

- 1 Graph the average time taken for the cross to disappear against the temperature of the sodium thiosulfate.
- 2 What happened to the time taken for the cross to disappear as the temperature increased?
- 3 What is the relationship between temperature and the rate of the reaction?
- 4 Do your results support the theory about the effect of temperature on the rate of reactions?

Discussion

- 1 Where might errors have occurred in the experiment and how would these have affected the results?
- 2 Were your results accurate and reliable? Explain why or why not.
- 3 How could you improve the experiment if you were to do it again?

Conclusion

What did you find out about the effect of temperature on the rate of the reaction between sodium thiosulfate and dilute hydrochloric acid?

Taking it further

How could you further investigate the effect of temperature on the rate of reactions?

ACTIVITY 9.9

DODGEM CAR CRASH ANALOGY – TEMPERATURE

Aim

To consider the effect of temperature on reaction rates in the dodgem car crash analogy

What to do

- 1 Review the dodgem car crash analogy in Activity 9.3.
- 2 In your analogy what would be the equivalent of temperature? (Remember that temperature measures the average kinetic energy.)
- 3 How would increasing this affect the rate of dodgem car crashes?
- 4 Is this consistent with the effect of temperature on reaction rates?

What did you discover?

Write a paragraph summarising how the equivalent of temperature will affect the rate of dodgem car crashes.

Catalysts

EFFECT OF A CATALYST ON THE DECOMPOSITION OF HYDROGEN PEROXIDE

View this animation to see the effect of a catalyst on the rate of decomposition of hydrogen peroxide, which catalyst and the amount of catalyst to add.

A **catalyst** is something that can affect the rate of a reaction without being consumed in the reaction. Positive catalysts increase the rate of the reaction. Negative catalysts decrease the rate of the reaction. Usually, we tend to refer to positive catalysts.

There are various ways that a catalyst can affect a reaction, but all positive catalysts provide an alternative pathway for the reaction, which is easier and therefore has a lower activation energy. A lower activation energy means that more molecules have enough energy for successful collisions. Therefore, there are more successful collisions and therefore a faster rate of reaction.

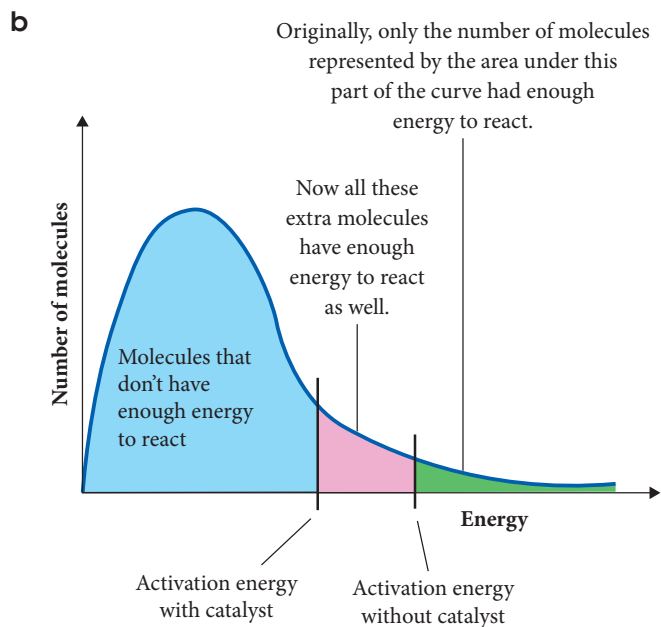
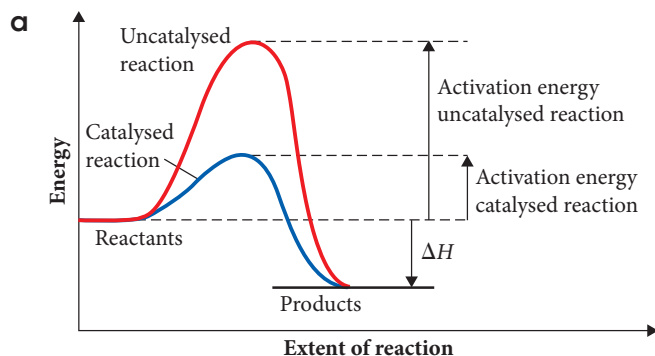


Figure C9.23 ►

The effect of a catalyst on the number of molecules with sufficient energy for a successful collision: a) energy profile diagram of a catalysed and an uncatalysed reaction; b) Maxwell-Boltzmann distribution for catalysed and uncatalysed reactions

Catalysts affect the rate of certain reactions by providing an alternative reaction pathway with a reduced activation energy.

One type of catalyst that is being used more commonly is **metal nanoparticles**. A nanoparticle is a particle with a diameter less than 100 nm that behaves as a whole unit. Metal nanoparticles have an advantage over other metal catalysts due to their large surface area. This means that a large amount of the reactants can access the catalyst and therefore they can have a greater impact on the reaction rate. Currently, scientists are conducting research into different metal nanoparticles for a number of reactions, including gold and copper nanoparticles as catalysts for fuel cells.

Catalysts in our bodies ensure that the reactions needed for a healthy life are occurring at the appropriate rate. Biological catalysts are called **enzymes** and are usually made of **protein**. Enzymes are affected by temperature and pH. As enzymes have to work within a living organism, it is logical that the optimal temperature and pH is that of the organism. Outside of these conditions, the enzyme is ineffective, and this can often mean that the reaction is not able to occur at the required rate.



For more information about enzymes, refer to Context 4, 'Making reactions work for us', page 85.

Cars are now equipped with catalytic converters. These structures are fitted so that the exhaust fumes pass over a layer of a metal catalyst, usually platinum, rhodium and/or palladium. As the gases pass over the catalyst, the harmful combustion products of nitrogen oxides, volatile organic compounds and carbon monoxide are converted to less harmful substances. Therefore, the catalytic converter is important in reducing pollution from our cars.

EXPERIMENT 9.6

EFFECT OF A CATALYST ON RATE OF REACTION

Oxalic acid reacts with acidified potassium permanganate to produce colourless products. The intense purple of the permanganate disappears as the reaction proceeds. The time for this to occur can be measured and used to compare the rate of reactions.

Manganese(II) sulfate can act as a catalyst for this reaction, increasing the rate of the reaction and therefore decreasing the time taken for the decolorisation to occur.

Aim

To observe the effect of a catalyst on the rate of the reaction between oxalic acid and acidified permanganate

Materials

- 6 mL of 0.02 mol L⁻¹ potassium permanganate (KMnO₄)
- 50 mL of 2 mol L⁻¹ sulfuric acid (H₂SO₄)
- 250 mL of distilled water
- 6 mL of oxalic acid (H₂C₂O₄)
- 1 mL of manganese(II) sulfate (MnSO₄)
- Stopwatch
- 2 × 250 mL beakers
- 3 × 10 mL, 1 × 25 mL and 1 × 250 mL measuring cylinders
- Stirring rod

What are the risks in doing this experiment?	How can you manage these risks to stay safe?
Chemicals could splash in your eyes.	Wear safety glasses at all times.
2 mol L ⁻¹ H ₂ SO ₄ is corrosive to skin and clothes.	Wear gloves and an apron.
KMnO ₄ is an oxidising acid; it can irritate the skin and eyes.	Wear gloves and an apron.
Oxalic acid is harmful to the skin and eyes.	Wear gloves and an apron.

In your write-up, add any more risks you can think of, as well as ways to manage them. In particular, expand on the risks listed to identify specific risks involved with each of them.

Procedure

- 1 Into the first beaker place 25 mL of 2 mol L⁻¹ H₂SO₄, 125 mL of distilled water, 3 mL of 0.02 mol L⁻¹ KMnO₄ and 3 mL of saturated oxalic acid.
- 2 Start the stopwatch immediately and record how long it takes for the solution to become colourless.
- 3 Into the second beaker place 25 mL of 2 mol L⁻¹ H₂SO₄, 125 mL of distilled water, 3 mL of 0.02 mol L⁻¹ KMnO₄, 3 mL of saturated oxalic acid and 1 mL of saturated MnSO₄.
- 4 Start the stopwatch immediately and record how long it takes for the solution to become colourless.
- 5 Follow your teacher's instructions regarding disposal of chemicals.

Results

Record your results in a table similar to the one below.

Beaker	Time taken for decolorisation (s)
No MnSO ₄	
With MnSO ₄	

Analysis of results

- 1 What was the effect of adding MnSO_4 to the reaction mixture?
- 2 Given that MnSO_4 was acting as a catalyst, did your results support the theory about the effect of catalysts on the rate of reactions?

Discussion

How could the experiment be improved to obtain more accurate and reliable data?

Conclusion

What did you determine about the effect of MnSO_4 as a catalyst on the reaction between oxalic acid and potassium permanganate?

Taking it further

- 1 How could you further investigate the effect of catalysts on the rate of reaction?
- 2 How could you determine if other factors influence the effect of catalysts on the rate of reactions?

ACTIVITY 9.10

DODGEM CAR CRASH ANALOGY – CATALYST

Aim

To consider the effect of a catalyst on reaction rates in the dodgem car crash analogy

What to do

- 1 Review the dodgem car crash analogy in Activity 9.3.
- 2 In your analogy, what would be the equivalent of a catalyst? (Remember that a catalyst will make it easier for a collision to be successful because less energy is needed for a successful collision.)
- 3 How would adding this catalyst affect the rate of dodgem car crashes?
- 4 Is this consistent with the effect of catalysts on reaction rates?

What did you discover?

Write a paragraph summarising how a catalyst will affect the rate of dodgem car crashes.

QUESTION SET 9.3

Remembering

- 1 List the factors that affect the rate of a reaction.
- 2 Define 'catalyst'.
- 3 Draw a Maxwell-Boltzmann distribution to show the distribution of energy of particles at two different temperatures. Clearly identify which curve is for the higher temperature.

Understanding

- 4 Explain how crushing a lump of solid will increase its surface area.
- 5 Compare the average kinetic energy of the particles of substances at 10°C and at 40°C .
- 6 Explain how the pressure of a gas is equivalent to its concentration.
- 7 Use collision theory to explain how each of the following factors affects the rate of a reaction.
 - a Concentration or pressure
 - b Temperature
 - c Catalyst
 - d Surface area

Applying

- 8 Ben and Aimee both added a sugar cube to their cups of tea, which were at the same temperature. Ben stirred his tea, while Aimee left her cup to sit. After 1 minute, they each drank their tea. Whose tea would have been sweeter? Justify your answer.

Analysing

- 9 Five test tubes each contained HCl and magnesium as shown in the following table. Which test tube would contain the fastest reaction? Justify your answer.

Test tube	Concentration of HCl (mol L ⁻¹)	Temperature (°C)	Size of 2 g of magnesium
1	1	25	Lump
2	2	25	Long strip
3	1	10	Strip cut into small pieces
4	0.5	40	Long strip
5	2	40	Strip cut into small pieces

Reflecting

- 10 Explain how the development of collision theory has enabled us to understand why reactions occur at different rates.

CHAPTER CHECKLIST

You should know:

- rate tells us how quickly one quantity changes compared to another quantity
- the slope (or gradient) of a graph of a reaction indicates the rate of the reaction
- to measure the rate of a reaction, we measure the changes in reactants or products in a certain time
- we can observe the rate of a reaction by changes in colour, mass, temperature or pH
- collision theory states that, for a reaction to occur, the particles need to collide with sufficient energy and correct orientation
- collision theory allows us to understand what happens during a reaction and, therefore, the reason for the rate of reactions
- activation energy is the amount of energy needed for a successful collision and is the amount of energy required to break the bonds in the reactants
- the correct orientation allows for the appropriate atoms to align so that new bonds can form
- the transition state is the moment when the reactants' bonds have broken but the products' bonds have not yet formed, and the substance is known as the activated complex. This is a very unstable, high-energy state that only lasts for a very short time
- a number of factors affect the number of successful collisions during a reaction, and therefore affect the rate of the reaction
- the concentration or pressure of the reactants affects the number of collisions between reactants. The higher the concentration of the reactants, the more collisions will occur. This means that there will be more successful collision in a given time and a faster rate of reaction

- dividing a solid increases the surface area available for collisions. This means that there are more collisions in total, more successful collisions in a given time and thus a faster rate of reaction
- when temperature increases, the average kinetic energy of particles increases. This means that more particles have enough energy for a successful collision. Therefore, there are more successful collisions in a given time and the reaction rate increases
- when temperature increases, the particles are also moving faster due to the greater average kinetic energy. This means that they will collide more frequently, and the number of successful collisions will be increased and thus the rate of the reaction will be increased. Although this effect is less significant than that due to kinetic energy, it does contribute to the effect on the reaction rate
- a catalyst is a substance that affects the rate of a reaction without being consumed. Most catalysts are positive catalysts because they increase the rate of the reaction by providing an alternative pathway for the reaction, which has a lower activation energy. This means that more particles will have enough energy for a successful collision, increasing the number of successful collisions in a given time and increasing the rate of the reaction.

You should be able to:

- represent the changes in energy during a reaction by an energy profile diagram.

CHAPTER GLOSSARY

activated complex the complex formed when the reactants' bonds are breaking but the products' bonds have not yet formed (see *transition state*)

activation energy the energy required to break the bonds of the reactants and therefore the minimum energy required for a successful collision

catalyst a substance that changes the rate of a reaction without being consumed

collision theory a theory that explains the rate of reaction at a molecular level; it states that, for a reaction, the particles must collide with sufficient energy and the correct orientation

concentration the amount of solute dissolved in a given volume of solvent

endothermic reaction a reaction that absorbs heat from the surroundings because the reactants have less enthalpy than the products

energy profile diagram a diagram showing the relative energies during the progress of a reaction

enthalpy the total energy possessed by a chemical substance at a constant pressure; it is usually expressed as the change in enthalpy

enzyme a protein molecule that catalyses a specific type of reaction by lowering the energy of activation

exothermic reaction a reaction that releases energy to the surroundings because the products have less energy than the reactants

gradient the slope or steepness of a line; it is calculated by dividing the vertical height (rise) by the horizontal distance (run)

kinetic energy the energy of movement

Maxwell-Boltzmann distribution a graph showing the distribution of energies (or velocities) of particles

metal nanoparticle a metal particle with a diameter of less than 100 nm that behaves as a whole unit

pressure the force exerted by a gas or gases due to the collisions of the particles with the walls of a container; expressed as force per unit area

protein a large organic molecule made up of many amino acids joined together

rate how much one quantity changes with respect to another quantity

surface area the total area of all surfaces

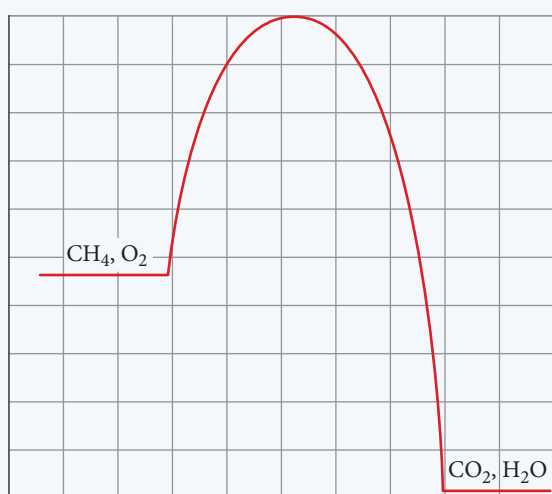
temperature a measure of the average kinetic energy of the particles

transition state the temporary, unstable state when the reactants are forming products (see *activated complex*)

CHAPTER REVIEW QUESTIONS

Remembering

- 1 Define:
 - a rate.
 - b surface area.
 - c concentration.
 - d pressure.
 - e catalyst.
- 2 List the factors that affect the rate of a reaction, and state how each factor achieves this.
- 3 How can a graph be used to determine the rate at different times during a reaction?
- 4 Draw a Maxwell-Boltzmann distribution.
- 5 Copy the energy profile diagram below and label the axes, reactants, products, activation energy and activated complex.



◀ **Figure C9.24**
Energy profile diagram for the combustion of methane

Understanding

- 6 Explain why particles must collide in order to react.
- 7 Draw an energy profile diagram for an endothermic reaction in which strong bonds are broken in the reactants. Describe the diagram you have drawn and the reasons for each aspect.
- 8 Why is the rate of a reaction fastest at the start of the reaction?
- 9 Explain how catalysts increase the rate of reactions.
- 10 Why does the activated complex have more energy than the reactant or products?

Applying

- 11 If someone is paid \$525.00 after working 6 hours each day for 5 days, what is their hourly rate of pay?
- 12 Sketch a graph showing the change in temperature during an exothermic reaction.
- 13 Sam noticed when she started using cold water to wash her clothes that the washing powder was not dissolving properly. Explain why this might happen.

- 14** In an experiment, some hydrogen peroxide was added to a reaction flask, which was connected to a gas syringe. A small amount of manganese(II) dioxide was added and a rapid reaction produced bubbles of a gas (oxygen gas), which was collected in the syringe. After 6 minutes, no more gas was produced.
- Sketch a graph for the volume of gas produced in the experiment.
 - When the reaction was complete, manganese dioxide was present. What was the purpose of the manganese dioxide?
 - Describe one way that you could make the reaction proceed more slowly. How would this affect the volume of gas collected?
- 15** The graph in Figure C9.25 shows the concentration of nitrogen dioxide (NO_2) gas produced during the reaction between NO and O_2 gas. Sketch the graph you would expect if more NO gas had been initially added to the container.

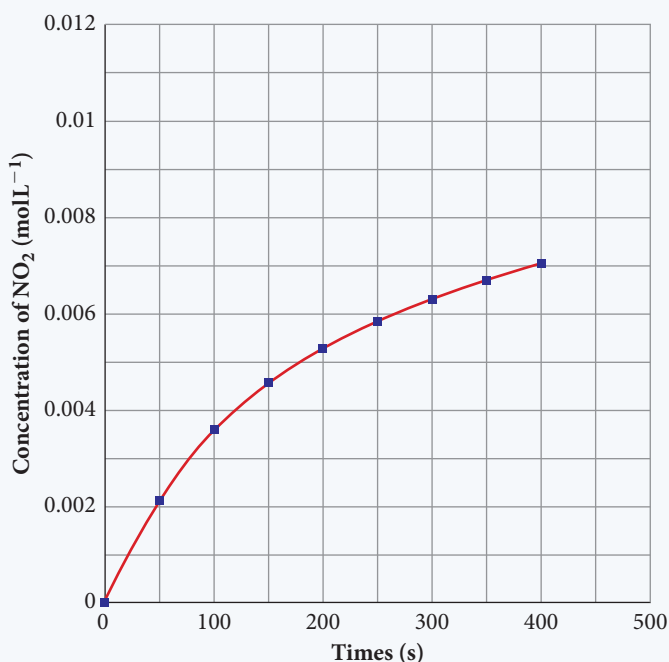


Figure C9.25
 NO_2 produced during the reaction between $\text{NO}(\text{g})$ and $\text{O}_2(\text{g})$

Analysing

- 16** An analogy of how enzymes work is the comparison with climbing a mountain to get to the other side. In this analogy, the reaction without an enzyme is compared to walking to the very top of the mountain and then back down the other side. The reaction with the enzyme is compared to walking through a tunnel in the mountain to get to the other side. Discuss the merits of this analogy.

Reflecting

- 17** Use your understanding of collision theory and the factors that affect the rate of a reaction to compare playing darts with a chemical reaction.
- 18** Draw a concept map to reflect your understanding of collision theory and the factors that affect the rate of a reaction.